

Dat Nguyen

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EDUCATION

University of Pennsylvania

Master of Science in Computer Science

Philadelphia, PA

Aug 2026 – May 2028

Michigan State University

Bachelor of Science in Computer Science, Minor in Business

East Lansing, MI

Aug 2022 – May 2026

WORK EXPERIENCE

Jack

Software Engineer Intern

New York, NY

May 2026 – Aug 2026

- Automated job discovery and application for **250K+** users by architecting a distributed pipeline (**Next.js**, **Supabase**, **pg-cron** RPC scheduling) that scrapes **12+** **ATS providers** and drives autonomous browser-based applications on candidates' behalf.
- Improved candidate-job match quality by training a **LightGBM** ranking model to predict interview conversion (leakage-safe splits, **PR-AUC**-optimized for a ~8% positive class), shipped as an env-gated re-ranking layer for safe progressive rollout alongside the production matching engine.
- Built an **LLM-driven** resume pipeline (**Azure OpenAI**) with deterministic, hash-based template routing across **5 LaTeX** templates and **LuaLaTeX/XeLaTeX** compilation, using fail-loud validation to guarantee reproducible, ATS-compliant PDFs.

DIJJAM

Software Engineer Intern

Remote

May 2025 – Aug 2025

- Enabled seamless subscription management for **1M+** users and **15K+** **DJs** by designing and building a scalable, distributed pricing-tier and payment platform in **Next.js/TypeScript** that handles recurring billing, upgrades, and cancellations.
- Helped **500+** **DJs** earn recurring revenue by designing and implementing a **Python/Flask** affiliate system with **PostgreSQL**-backed commission-tracking algorithms and automated payout scheduling.
- Saved **8+** hours/week of manual bookkeeping by automating invoice and payment reconciliation using the **Gmail API**, applying string-matching algorithms to parse transaction emails and reconcile them against subscription records.

nuoc.solutions

Software Engineering Intern

San Francisco, CA

May 2024 – May 2025

- Delivered real-time water-quality insights to **1,000+** farmers by designing and building a scalable, map-based monitoring platform (**Next.js**, **FastAPI**, **PostgreSQL**) covering **200+** stations.
- Enabled **500+** monthly natural-language irrigation queries by developing an LLM-powered information-retrieval system over historical and seasonal time-series data for non-technical users.
- Scaled the platform to support multi-province citizen reporting by containerizing services with **Docker** and deploying on **Azure** with horizontal autoscaling in a **Unix/Linux** production environment.

HCLTech

Software Engineering Intern

Hanoi, Vietnam

May 2023 – Aug 2023

- Cut manual security-review time by **70%** by designing and building a **GitHub Actions** CI/CD pipeline that runs automated static-analysis (SAST) scans across **10K+** files on every commit.
- Flagged **150+** high-risk security events at **95% precision** by developing and training anomaly-detection algorithms (**Isolation Forest**, **One-Class SVM**) on large-scale event-log data.
- Accelerated detection of and response to security incidents by building real-time file-integrity monitoring infrastructure across **200+** production **Unix/Linux** systems.

PROJECTS

ACASO | Founder & Lead Engineer | FastAPI, Next.js, PostgreSQL, Cerebras, Groq, Fireworks AI

Mar 2026 – Present

- Reduced manual invoice-review time by **70%** and processing costs by **20%** for **5 client companies** by founding and leading development of an AI-powered B2B SaaS platform that matches, audits, and reconciles invoices against contract terms.
- Led a distributed team of **4 engineers**, collaborating directly with users to prioritize requirements, design system architecture, and ship production features on a large-scale codebase.
- Architected a low-latency, distributed inference pipeline across **Cerebras**, **Groq**, and **Fireworks AI** with cost/latency-aware routing algorithms, backed by a **FastAPI/PostgreSQL** real-time data-processing system.

Real-Time Crypto Trading Platform | Spring WebFlux, Redis Streams, PostgreSQL, Docker

- Supported high-concurrency, distributed trading workloads by architecting a low-latency, event-driven backend that ingests and processes live **WebSocket** market data via **Spring WebFlux** and **Redis Streams**.
- Shipped a cloud-ready, JWT-secured **REST API** on a **Unix/Linux**-based, fully containerized **Docker** deployment with automated integration tests for rapid, reliable iteration.

Modular Simulation Backend | C++, Qt, WebAssembly, Docker, GitHub Actions

- Built a high-performance, modular **C++** backend for distributed agent orchestration and real-time analytics, with **Docker/GitHub Actions** CI/CD across native (**Qt**) and **WebAssembly** targets.

TECHNICAL SKILLS

Languages: Java, Python, C++, TypeScript, JavaScript, SQL, Go, Swift, HTML/CSS

Frameworks & Tools: Spring Boot, Spring WebFlux, Next.js, React, Node.js, FastAPI, Flask, Docker, Kubernetes, AWS, Azure, PostgreSQL, Redis, Kafka, Terraform, Git, GitHub Actions, PyTorch, scikit-learn